ORAL ANATOMY AND OCCLUSION

1. Which tooth numbering system uses a two-digit binomial format where the first number represents the quadrant and the second represents the tooth?
   A. Universal Numbering System
   B. Palmer's Notation
   C. FDI System
   D. Zsigmondy System

2. What are the growth centers of a tooth called?
   A. Cusp
   B. Lobe
   C. Tubercle
   D. Cingulum

3. A convexity located specifically on the lingual cervical third of an anterior tooth is known as the:
   A. Mamelon
   B. Cingulum
   C. Tubercle
   D. Ridge

4. Rounded protuberances found on the incisal ridges of newly erupted incisors are called:
   A. Ridges
   B. Cysts
   C. Mamelons
   D. Sulci

5. Which type of depression is defined as a long depression between ridges and cusps?
   A. Fossa
   B. Pit
   C. Sulcus
   D. Groove

6. The geometric outline of all anterior teeth from the mesial/distal (MD) aspect is:
   A. Trapezoidal
   B. Triangular
   C. Rhomboidal
   D. Ovoid

7. What is the geometric outline of mandibular posterior teeth from the mesial/distal (MD) aspect?
   A. Triangular
   B. Rhomboidal
   C. Trapezoidal
   D. Diamond-shaped

8. Which tooth numbering system has been officially recognized and adopted by the World Health Organization (WHO)?
   A. Palmer's Notation
   B. Zsigmondy System
   C. FDI System
   D. Universal Numbering System

9. Which teeth share a contact area located at the middle third (M) on both their mesial and distal surfaces?
   A. Maxillary Centrals and Laterals
   B. Mandibular Centrals and Laterals
   C. Maxillary Premolars and Molars
   D. Mandibular Premolars and Molars

10. The largest occlusal embrasure in the permanent dentition is found between which two teeth?
    A. Maxillary Centrals
    B. Maxillary Lateral and Maxillary Canine
    C. Maxillary Canine and Maxillary 1st Premolar
    D. Mandibular Centrals

11. Where are the smallest incisal embrasures located?
    A. Between Maxillary Centrals
    B. Between Maxillary Laterals and Canines
    C. Between Mandibular Centrals
    D. Between Mandibular Laterals and Canines

12. What is the location of the facial height of contour for all anterior teeth?
    A. Cervical third
    B. Middle third
    C. Incisal third
    D. Junction of cervical and middle third

13. For mandibular molars, where is the facial height of contour located?
    A. Cervical third
    B. Middle third
    C. Junction of cervical and middle third
    D. Occlusal third

14. Which specific tooth features a lingual height of contour located at the occlusal third?
    A. Maxillary 1st Premolar
    B. Mandibular 1st Premolar
    C. Mandibular 2nd Premolar
    D. Maxillary 1st Molar

15. On which tooth and surface is the greatest line of CEJ curvature (contour) found?
    A. Distal of Mandibular Central Incisor
    B. Mesial of Maxillary Central Incisor
    C. Mesial of Maxillary Canine
    D. Distal of Maxillary Central Incisor

16. How do the mesial and distal CEJ lines curve on a tooth crown surface?
    A. Apically
    B. Coronally
    C. Horizontally straight
    D. Towards the lingual

17. What forms the base of the triangular interproximal space between adjacent teeth?
    A. Proximal contact area
    B. Alveolar process
    C. Marginal ridges
    D. Interdental papilla

18. Which permanent anterior tooth has the widest mesiodistal (MD) width?
    A. Maxillary Central Incisor
    B. Maxillary Canine
    C. Mandibular Central Incisor
    D. Mandibular Canine

19. Which tooth is characterized by having a shovel-shaped lingual surface, peg-shaped variations, and a palatogingival groove?
A. Maxillary Central Incisor
B. Maxillary Lateral Incisor
C. Mandibular Lateral Incisor
D. Maxillary Canine

20. What feature distinguishes the crown of a maxillary canine from other anteriors?
A. Longest root trunk
B. Widest anterior tooth faciolingually (FL)
C. Straightest mesial edge
D. Bifurcated roots

21. Which tooth is considered the "cornerstone of the mouth" and the most stable tooth?
A. Maxillary Central Incisor
B. Maxillary Canine
C. Mandibular 1st Molar
D. Maxillary 1st Molar

22. The mesial cusp ridge is longer than the distal cusp ridge (MCR > DCR) on which permanent tooth?
A. Maxillary Canine
B. Maxillary 1st Premolar
C. Mandibular 1st Premolar
D. Mandibular Canine



23. Which is the only permanent premolar that typically possesses two roots (buccal and palatal)?
A. Maxillary 1st Premolar
B. Maxillary 2nd Premolar
C. Mandibular 1st Premolar
D. Mandibular 2nd Premolar

24. A marked mesial developmental depression/ concavity on the crown and root is a prominent feature of which tooth?
A. Maxillary Central Incisor
B. Maxillary 1st Premolar
C. Maxillary 2nd Premolar
D. Mandibular 1st Premolar

25. Compared to the maxillary 1st premolar, the maxillary 2nd premolar has:
A. A longer central groove and more supplemental grooves
B. A more wrinkled appearance with a shorter central groove and equal cusp heights
C. Two distinct roots and a marked mesial concavity
D. A longer mesial cusp ridge than distal cusp ridge



26. Which is the largest permanent tooth in the maxillary arch and possesses the widest faciolingual (BL) dimension?
A. Maxillary Canine
B. Maxillary 1st Premolar
C. Maxillary 1st Molar
D. Maxillary 2nd Molar

27. The Cusp of Carabelli is a non-functional fifth cusp located on which cusp of the maxillary 1st molar?
A. Mesiobuccal (MB)
B. Distobuccal (DB)
C. Mesiolingual (ML)
D. Distolingual (DL)

28. What is the correct order of cusp heights on the maxillary 1st molar, from tallest to shortest?
A. MB > ML > DB > DL
B. ML > MB > DB > DL
C. DL > DB > MB > ML
D. ML > DB > MB > DL



29. Which root of the maxillary 1st molar is the longest?
A. Mesiobuccal root
B. Distobuccal root
C. Palatal root
D. Distal root

30. The crown of a maxillary 2nd molar frequently exhibits an occlusal outline described as:
A. Hexagonal
B. Square
C. Rhomboidal or Heart-shaped
D. Triangular

31. Which permanent maxillary tooth is located closest to the opening of Stensen's duct?
A. Maxillary 1st Premolar
B. Maxillary 1st Molar
C. Maxillary 2nd Molar
D. Maxillary 3rd Molar

32. What is the smallest tooth in the permanent human dentition?
A. Mandibular Central Incisor
B. Mandibular Lateral Incisor
C. Maxillary Lateral Incisor
D. Mandibular 1st Premolar

33. Which permanent tooth normally occludes with ONLY ONE opposing tooth?
A. Maxillary Central Incisor
B. Mandibular Central Incisor
C. Maxillary Canine
D. Mandibular 2nd Molar

34. A distinct feature of the mandibular lateral incisor crown when viewed from the incisal aspect is its:
A. Symmetrical triangular outline
B. Distolingual twist
C. Centrally located cingulum
D. Prominent labial ridge

35. Which variation in root anatomy is sometimes observed in mandibular canines?
A. Trifurcated roots
B. Bifurcated roots (facial and lingual)
C. Fused roots with three canals
D. Hooked palatal roots

36. Which tooth is the smallest among all premolars and exhibits a "snake eye" appearance on its occlusal surface?
A. Maxillary 1st Premolar
B. Maxillary 2nd Premolar
C. Mandibular 1st Premolar
D. Mandibular 2nd Premolar

37. The mesiolingual developmental groove is a characteristic landmark of which tooth?
A. Maxillary 1st Premolar
B. Mandibular 1st Premolar
C. Mandibular 2nd Premolar
D. Mandibular 1st Molar

38. The most common occlusal configuration of the mandibular 2nd premolar is the:
A. H-type (2 cusps)
B. U-type (2 cusps)
C. Y-type (3 cusps)
D. Diamond-type (1 cusp)

39. The apex of the root of a mandibular 2nd premolar lies in close proximity to which anatomical structure?
A. Stensen's duct
B. Maxillary sinus
C. Mental foramen
D. Incisive foramen

40. What is the largest tooth in the mandibular arch and widest mesiodistally?
A. Mandibular Canine
B. Mandibular 1st Molar
C. Mandibular 2nd Molar
D. Mandibular 3rd Molar

41. How many cusps does a typical mandibular 1st molar possess?
A. 3
B. 4
C. 5
D. 6

42. What is the correct relative cusp size order for a mandibular 1st molar?
A. MB > ML > DL > DB > Distal
B. ML > MB > DB > DL > Distal
C. Distal > DB > DL > MB > ML
D. MB > DB > ML > DL > Distal



43. Which permanent molar has a rectangular occlusal outline, 4 cusps of equal height, and a straight central groove with no distal cusp?
A. Maxillary 1st Molar
B. Mandibular 1st Molar
C. Mandibular 2nd Molar
D. Mandibular 3rd Molar

44. Which permanent tooth is most frequently impacted and also most commonly missing congenitally?
A. Maxillary Canine
B. Maxillary 1st Premolar
C. Mandibular 2nd Premolar
D. Mandibular 3rd Molar

45. Compared to permanent teeth, primary anterior crowns are:
A. Longer cervicoincisally than mesiodistally
B. Wider mesiodistally than their crown length
C. Thicker in enamel and dentin
D. Darker and more pigmented

46. What feature is characteristic of primary roots compared to permanent roots?
A. Primary posterior roots are shorter and thicker
B. Primary posterior roots are longer, more slender, and flared
C. Primary roots have smaller pulp canals
D. Primary roots lack cervical constriction

47. Primary pulp chambers and pulp horns are characterized as being:
A. Small chambers with low pulp horns
B. Large chambers with high pulp horns
C. Obliterated by secondary dentin
D. Identical in size to permanent teeth

48. The primary maxillary canine features a unique incisal ridge relationship where the:
A. Mesial cusp ridge is longer than the distal cusp ridge
B. Distal cusp ridge is longer than the mesial cusp ridge
C. Cusp tip is displaced to the distal
D. Cusp is completely blunted

49. Which primary tooth has a crown that resembles a maxillary 1st premolar and features an H-shaped cuspal configuration?
A. Primary Maxillary Central Incisor
B. Primary Maxillary 1st Molar
C. Primary Maxillary 2nd Molar
D. Primary Mandibular 1st Molar

50. The primary maxillary 2nd molar morphologically resembles which permanent tooth?
A. Maxillary 1st Premolar
B. Maxillary 2nd Premolar
C. Maxillary 1st Molar
D. Maxillary 2nd Molar

51. Which primary tooth is the widest buccolingually (BL) among all primary teeth?
A. Primary Maxillary 1st Molar
B. Primary Maxillary 2nd Molar
C. Primary Mandibular 1st Molar
D. Primary Mandibular 2nd Molar

52. Which primary tooth is the widest mesiodistally (MD) among all primary teeth and resembles the permanent mandibular 1st molar?
A. Primary Maxillary 2nd Molar
B. Primary Mandibular 1st Molar
C. Primary Mandibular 2nd Molar
D. Primary Mandibular Canine

53. Which cusps act as the working (supporting) cusps in normal occlusion?
A. Maxillary Buccal and Mandibular Lingual
B. Maxillary Lingual and Mandibular Buccal
C. Maxillary Buccal and Mandibular Buccal
D. Maxillary Lingual and Mandibular Lingual

54. Non-working (guiding) cusps contact which areas of the opposing teeth?
A. Marginal ridges and fossae
B. Central pits only
C. Grooves and embrasures
D. Cingulum and incisal edges

55. How many "Keys to Occlusion" were established?
A. 4
B. 5
C. 6
D. 8

MR Molar Rel
CA Crown Angulation
CI Crown Inclination
Cu Curve of Spee
R Rotation
T Tight Contact

56. The postural position of the mandible is best described as a:
A. Tooth-guided position with maximum intercuspation
B. Muscle-guided position where no tooth contact occurs
C. Ligament-guided position at the most superior condylar location
D. Bone-guided position during deglutition

57. Centric Relation (CR) is defined as a:
A. Tooth-guided position
B. Muscle-guided rest position
C. Ligament-guided position
D. Purely translational movement of the condyles

58. What happens to posterior teeth during protrusive mandibular movement in normal occlusion?
A. They disarticulate (disocclude)
B. They maintain maximum intercuspation
C. Working cusps contact non-working cusps
D. They undergo heavy lateral force

59. Maximum posterior disarticulation during protrusion is achieved when anterior teeth are in an:
A. Overjet of 5 mm
B. Edge-to-edge relationship
C. Deep overbite position
D. Open bite position

60. During lateral mandibular movement, the side toward which the mandible moves is known as the:
A. Non-working side
B. Balancing side
C. Working side
D. Protrusive side

61. Canine-guided occlusion results in:
A. Contact of all posterior teeth during lateral excursions
B. Disocclusion of posterior teeth via overlap of maxillary and mandibular canines
C. Group function on the non-working side
D. Equal contact across all incisors and premolars

62. Group function occlusion is also referred to as:
A. Canine protection
B. Mutual protection
C. Unilateral balanced occlusion
D. Bilateral balanced occlusion

63. The Bonwill equilateral triangle connects which three anatomical points?
A. Incisal edges of central incisors and both mental foramina
B. Both condyles and the mandibular incisal point
C. Both maxillary canines and the mesial cusp of the 1st molar
D. Cusp tips of both canines and the anterior nasal spine

64. The Curve of Spee describes an arch curvature in which plane?
A. Medio-lateral
B. Antero-posterior
C. Vertical-transverse
D. 3D Spherical

65. The Curve of Wilson represents curvature in which plane?
A. Antero-posterior
B. Medio-lateral
C. Coronal-sagittal
D. 3D Spherical

66. Which curvature combines both the Curve of Spee and the Curve of Wilson into a 3D alignment?
A. Curve of Bonwill
B. Curve of Monson
C. Curve of Zsigmondy
D. Curve of Carabelli

67. Where does carbohydrate digestion first begin in the stomatognathic system?
A. Stomach during deglutition
B. Esophagus during peristalsis
C. Mouth during mastication
D. Pharynx during swallowing

68. What structural actions prevent food from entering the airway during deglutition?
A. Opening the glottis and epiglottis
B. Shutting the epiglottis and nasopharynx
C. Contracting the masseter and temporalis
D. Retracting the tongue into the sublingual space

69. Which primary molar has a rhomboidal occlusal shape and exhibits an oblique ridge?
A. Primary Maxillary 1st Molar
B. Primary Mandibular 1st Molar
C. Primary Mandibular 2nd Molar
D. Primary Maxillary 2nd Molar

70. Receptors providing sensory and proprioceptive input (such as taste, pressure, and position) are located in the muscles, TMJ, and:
A. Enamel matrix
B. Dental pulp chamber
C. Periodontal ligament
D. Cementum layer

primary teeth
MX
1M - rectangular
2M - rhomboidal

Md
1M - rhomboidal
2M - rectangular